

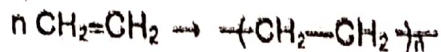
Polymers:

Indefinite combination of very large number of small molecules to form a large molecule is known as polymerization. The large molecule is called a polymer and the small molecule from which polymers are formed are called monomers.

Polymers are of two types 1. Addition polymerization 2. Condensation polymerization.

1. **Addition polymerization or Chain growth polymerization:** Simple molecules containing double or triple bonds join together by breaking up of the multiple bonds to form long chain polymers is called addition polymerization. Polymerization is initiated by heat, light, pressure or a catalyst. There is no liberation of small molecules during addition polymerization. Examples are.

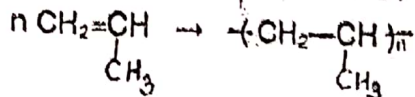
a) **Polyethene (PE):** Polymerization of ethylene at high temperature and pressure.



By using free radical initiator, low density LDPE and ionic catalyst high density HDPE are obtained.

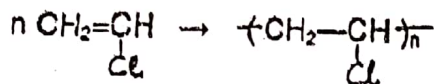
Properties: PE is resistant to strong acids and alkalis at room temperature. It is a thermoplastic and can be moulded to any shape, soluble in organic solvents. **Uses:** PE can be used for electric insulation, polythene bags, in moulded articles etc.

b). **Polypropene (PP):** Obtained by treating propylene with Ziegler-Natta catalyst ($\text{TiCl}_4 + \text{R}_3\text{Al}$)



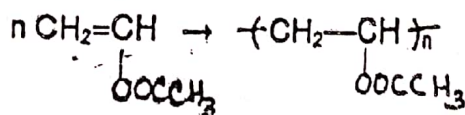
It is harder than polyethene, resistant to boiling water, has higher melting point. It is used for making disposable syringes, sterilizable hospital equipments, body of washing machines, wool like fibers, ropes, carpets, blankets etc.

c). **Polyvinylchloride (PVC)** formed from vinyl chloride.



Properties: It is colourless, non-inflammable, chemically inert, powder. It is resistant to light, inorganic acids, alkalis, soluble in organic solvents and brittle. **Uses:** It is used for making plastic pipes, rain coats, metal coatings, tank linings, safety helmets, refrigerator components, tyres, cycle and motorcycle mud guards etc.

d). **Polyvinyl acetate (PVA)** Prepared by polymerization of vinyl acetate in presence of peroxides.



It is a transparent, amorphous polymer, soluble in organic solvents. It is used to make water bases emulsion paints, textile fibres, leather, chewing gums etc.

2. **Condensation polymerization or Step growth polymerization.**

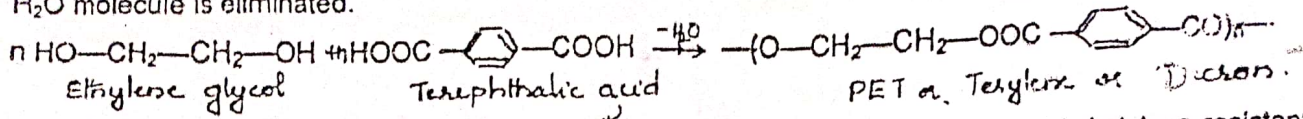
Two or more polyfunctional monomers (molecules containing more than one functional group) react together to form polymers with the elimination of simple molecules like H_2O , HCl , Methanol, NH_3 etc.

* Free radical mech
in addition

condensation of polyhydric alcohols with dicarboxylic acids.

PET (Polyethylene terephthalate). Formed by the polymerization of ethylene glycol and Terephthalic acid.

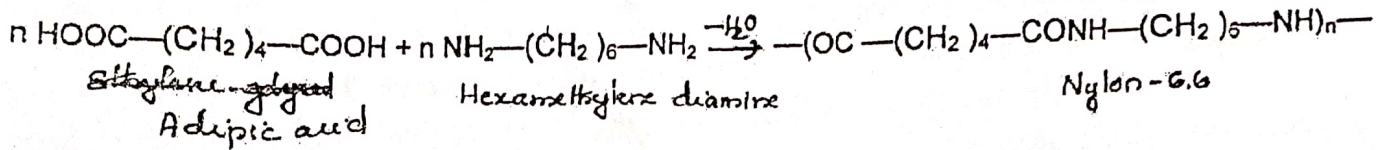
H₂O molecule is eliminated.



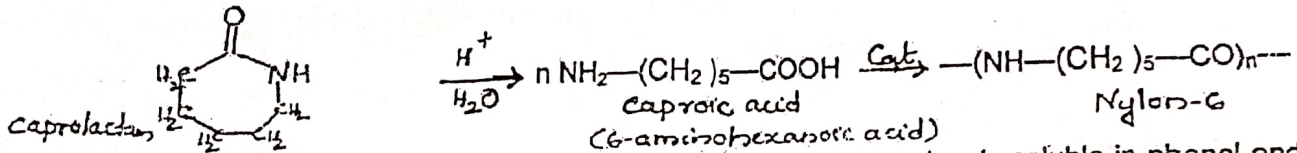
Properties: good fibre forming material, highly resistant to mineral and organic acids, but less resistant to alkalis and have high stretch resistance. **Uses:** Polyester threads are used in dress materials and home furnishings, PET bottles and containers, films, tarpaulin, film insulation and so on. Blendable with cotton and wool which enhances the life of cotton and woolen clothes.

Nylon is used as a general name for all fibre forming polyamides having protein-like structure.

Nylon-6,6. Is formed from Adipic acid and hexamethylene diamine in the presence of catalyst.



Nylon-6. Is made by the polymerization of Caprolactam in presence of a catalyst.



Properties of Nylons: Nylons are light polymers, insoluble in common solvents, soluble in phenol and formic acid, absorbs little moisture, very flexible and can be blended with wool.

Uses: Nylon-6,6. Is used as fibres for making garments, socks, carpets etc. Nylon-6 is used for making ropes, nylon bearings, tyre-cords, films etc.

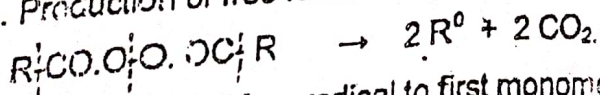
Mechanism of polymerization.

Four mechanisms are proposed for polymerization of ethylene and substituted compounds. Polymerization take place through three principal stages. 1). Chain initiation step 2). Chain propagation step and 3. Chain termination step.

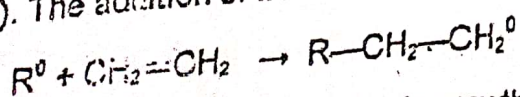
1. **Free radical mechanism:** Polymerisation by catalysts such as benzoyl peroxide, H₂O₂ initiates free radical mechanism. These catalysts are called free radical generators.

1). **Chain Initiation step:** Two reactions are involved in chain initiation step.

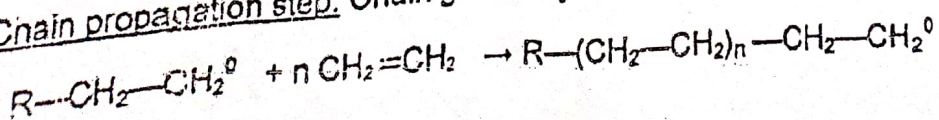
a). Production of free radicals by the homolytic fission of catalyst (chain initiator).



b). The addition of free radical to first monomer to produce chain initiator species.



2). **Chain propagation step:** Chain growth by successive addition of large numbers of monomer molecules.



Natural rubber is an elastomer obtained as latex from the bark of rubber trees. The milky white latex is collected in a pan and coagulated by agitating with formic acid or acetic acid. The coagulated rubber is squeezed by means of rollers, washed, dried and smoked to prevent fungal attack. Natural rubber is polyisoprene. Its molecular weight is of the order 300000.



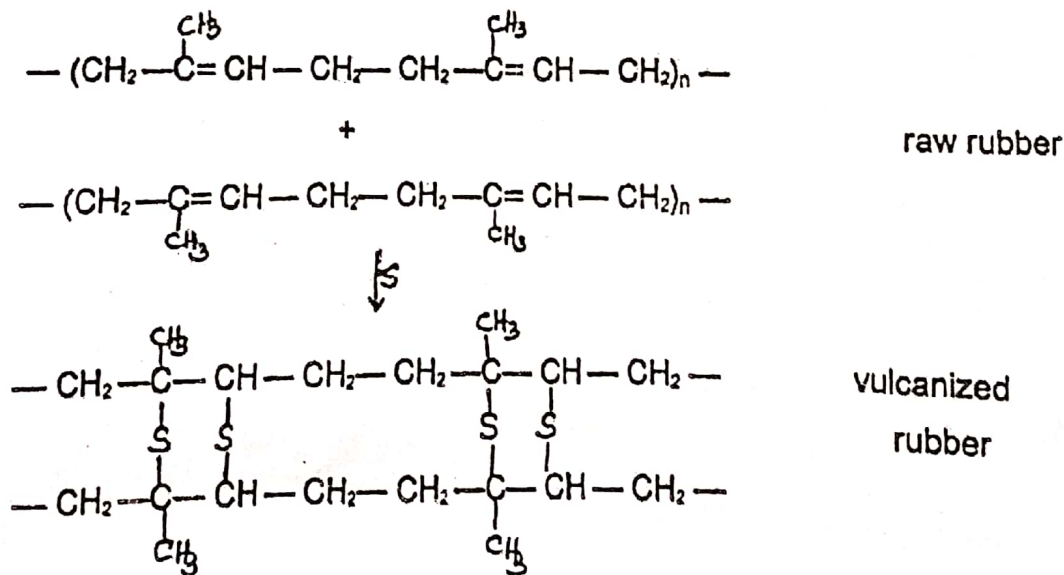
Rubber is elastic due to the coiled form of rubber chain. This straightens on stretching and on releasing the chain reverts to original coiled form.

Properties: Natural rubber is soft, sticky and insoluble in water, alcohol, dil. acids and alkalies, soluble in CS_2 , CCl_4 , petrol and turpentine. It has low tensile strength.

Drawbacks of raw rubber: 1) It is plastic in nature. 2) It becomes soft at high temperature and brittle at low temperature. So it can be used in 10 to 60°C. 3) It has large water absorption capacity. 4) Oxidized by HNO_3 , $\text{Con. H}_2\text{SO}_4$. 5) It degrades on exposure to air. 6) It swells in organic solvents and degrades. 7) It possesses tackiness and less durability.

Uses. Natural rubber is used to make erasers, elastic threads, balloons etc.

Vulcanization of rubber: Rubber is heated with 5 to 8% sulphur, 5% ZnO as filler, and 0.5% nitrogen containing compound as accelerator at 400 to 440K for half an hour. By vulcanization, double bonds in rubber are cross linked with sulphur between chains. Vulcanized rubber has better resistance to moisture, oxidation and low elasticity.



Advantage of vulcanization: 1) Prevent slippage 2) Make it less sensitive to temperature. useful in the range -40° to 100°C. 3) has only slight tackiness 4) low elasticity 5) Less water absorption capacity 6) resistant to organic solvents 7) High resistance wear and tear. 8) recovers original shape after the deforming load is removed. 9) easy to produce articles of desired shapes. 10. Better electric insulator.

Polymer Processing methods

Conversion of a polymer into various commercial products is called polymer processing. Polymer processing consists of three stages.

The plasticization stage

The solid polymer in the form of powder or granules is uniformly melted without decomposing the polymer, additives are added and mixed thoroughly. This process is called adding or compounding.

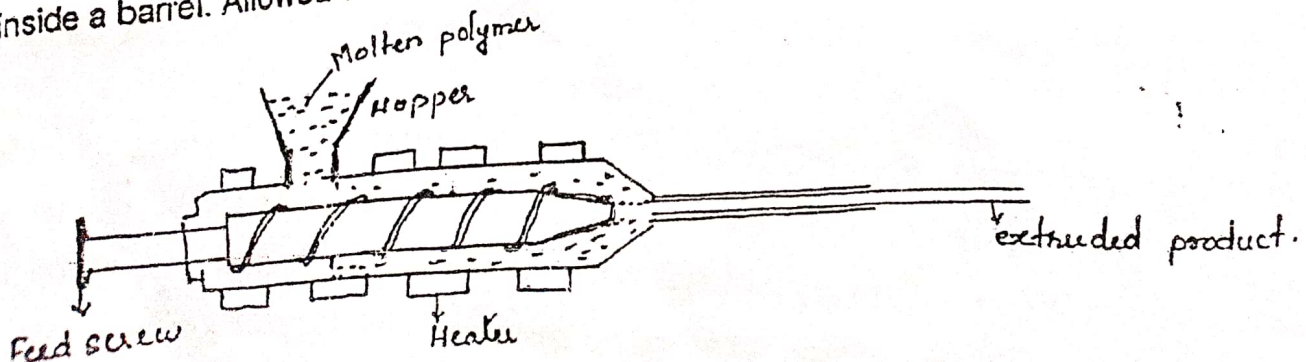
Compounding of plastics

The process of mixing the plastics with various other substances to improve the workability, flexibility, and to make it colourful, is called compounding. The main types of compounding ingredients and their functions are given below.

- ✓ **Catalysts or accelerators** - are added to accelerate the polymerization usually for thermosets. Usual catalysts are H_2O_2 , benzoyl peroxide, acetyl sulphuric acid, ZnO , metals like Na, Ag, Cu, Pb etc.
 - ✓ **Plasticizers** - are added to increase the plasticity and flexibility. The most commonly used plasticizers are vegetable oils, organophosphates, esters etc.
 - ✓ **Fillers** - are added to give better hardness, tensile strength, workability and finish. Some examples are Asbestos powder, carborundum, quartz, mica, wood flour, china clay, gypsum, marble powder, metal powder, cotton threads, scraps of clothes etc
 - ✓ **Lubricants** - like waxes, oils and soaps are added to impart glossy finish and antisticking property to plastics.
 - ✓ **Colouring materials** such as organic dyes and inorganic pigments are added to get a better colour to plastics.
 - ✓ **Stabilizers** - are added to improve the thermal stability during moulding process. Lead, leadchromate, red lead, stearates of lead. Cadmium, barium etc are used as stabilizers.
 - ✓ **Resin** - is a binder which holds the different constituents together
- II. **Casting stage** : The molten liquid polymer is moulded into desired shape.
- III. **The cooling stage** : cooling is done by spraying water or by circulating air by which the liquid sets into desired shape.

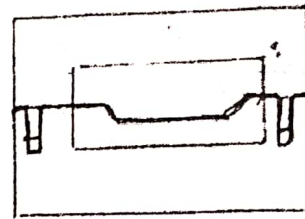
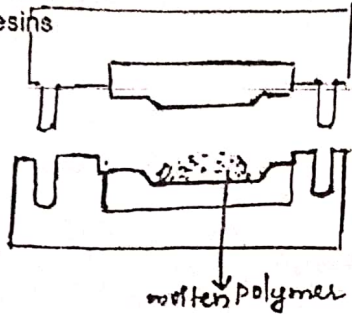
II. Moulding Techniques

1. **Extrusion moulding**: This method is used for the production of objects of uniform cross section like pipes, tubes, fibers, electric cables etc
The molten polymer liquid is forced into the die under pressure by means of a metal screw conveyor inside a barrel. Allowed to cool and set after moving from the die and cut into suitable length.



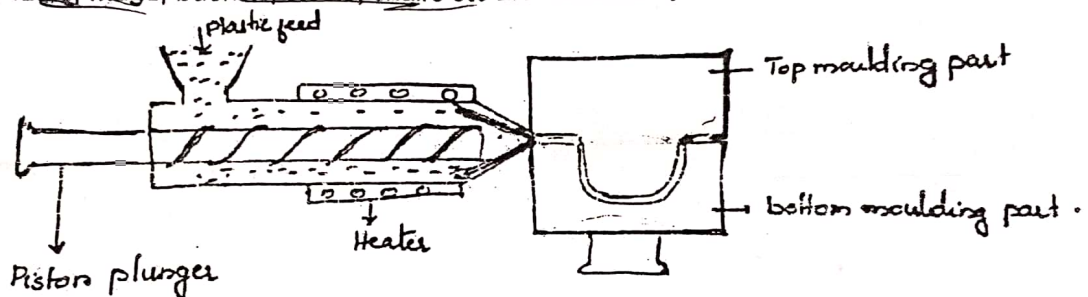
Compression moulding: This method is used to mould large parts which are flat or smoothly curved articles like fiber glass body parts for automobiles, boats, electrical switch panels etc.

Large sheets of thermoplastic polymer material are pre-heated and placed over one half of the mould and the other half of the mould is pressed over it so that the soft polymer material spreads and take up the desired shape of the mould. After moulding and cooling article is taken out by opening the mould. This is a low cost method where the wastage is minimum and can be applied to both thermoplastic and thermosetting resins.

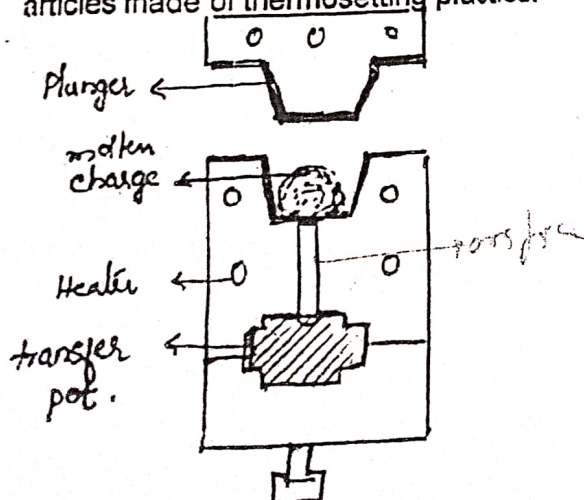


Mould closes

3. **Injection moulding:** This method is used to manufacture all types of plastic products with complicated shapes but one piece at a time. The molten polymer from the heating cylinder is injected at high pressure at a controlled rate through a small hole using a piston plunger or a screw arrangement into the suitable mould made of steel or aluminium. It is then allowed to cool. The mould can be opened up to take the finished product out. This method is widely used because of high speed, low mould cost, low loss of material and low finishing cost. Plates, mugs, buckets, stools, chairs etc are moulded by this method.



4. **Transfer moulding:** Pre weighed amount of polymer powder is heated in a separate chamber called transfer pot and then forced into a pre heated mould through an orifice into the mould cavity by a plunger at high pressure. It cures and sets in the cavity and then ejected out. This method is mainly used for moulding articles made of thermosetting plastics.



- * mechanism of lubrication
- * viscosity index
- * Flash point
- * Aniline point

Elimination of refractories medium: